Blood flow restriction: methods and apparatus still matter

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OVERVIEW OF BLOOD FLOW RESTRICTION APPLICATIONS

Application of 'blood flow restriction' (BFR) to the limbs at rest, during exercise or in combination with adjunct interventions (eg, electrical muscle stimulation, whole-body vibration) has emerged as a technique to improve health, performance, rehabilitation and recovery outcomes in healthy, clinical and high-performing populations (eg, military personnel, athletes and astronauts). This technique typically uses a pneumatic tourniquet cuff apparatus which is applied proximally on the limb(s), that includes an internal bladder system which is inflated with air to a predetermined pressure to compress the limb vasculature. The goal is to partially restrict arterial inflow and completely restrict venous outflow in tissues distal to the tourniquet cuff, creating a hypoxic environment that stimulates adaptive responses.[1] Other non-pneumatic BFR methods using less expensive straps have also been studied. While these approaches are more affordable and thus potentially address disparities in accessibility of BFR, evidence suggests that they pose significant challenges in ensuring the reliable and safe application of pressure, as discussed elsewhere (online supplemental 1). Applying BFR during exercise characterised by low external load alters local and systemic tissue physiology, exacerbating acute psychophysiological responses[2] and triggering a cascade of mechanistic pathways which, when applied repeatedly as a training intervention, elicits superior adaptations in the muscle, cardiopulmonary and vascular systems when compared with external load- or intensity-matched exercise. The aim of this manuscript is to outline evidence-based reporting recommendations for the design and publication of scientific data associated with BFR applications.

IMPACT OF BFR APPARATUS AND METHODOLOGIES ON PSYCHOPHYSIOLOGICAL RESPONSES

It is now well established that several key methodological variables impact the stimulus provided by BFR and, therefore, the subsequent psychophysiological responses.[2] For example, using tourniquet cuffs of narrow width requires higher absolute pressures to completely restrict arterial blood flow compared with wider cuffs,[3] and higher absolute and relative applied pressures exacerbate the psychophysiological responses to BFR exercise.[4] Such advancements in knowledge have naturally led to the development of guidelines for optimising BFR application; for example, if aiming to use lower absolute pressures, it is now recommended to use wider tourniquet cuffs and prescribe tourniquet cuff pressure relative to the individual's limb occlusion pressure, defined as the minimum pressure required to completely restrict arterial blood flow in the individual's limb(s), to minimise the perceptual and haemodynamic responses and maximise the safety and efficacy of BFR applications.[1]

Inevitably, the growing interest and adoption of BFR applications in research and practice has led to an increase in the number of commercially available BFR systems that offer new features and apparatus design, for example, autoregulation capabilities, different types of bladder systems (single vs multichambered) and bladder lengths.[5] Subsequent research indicates that these features can also impact the BFR stimulus provided; for example, Hughes *et al*[6] demonstrated differences between the set pressure (ie, the pressure to which the tourniquet cuff bladder is inflated) and the interface pressure (ie, the pressure between the tourniquet cuff and the limb) during BFR application at rest and during exercise, suggesting that transference of pressure from the cuff bladder to the limb is impacted by cuff material, thickness and shape. The compliance and stiffness of the cuff material may also influence how evenly the pressure is distributed and can influence blood flow velocity during passive application.[7] Though such tourniquet cuff designs and features have received less attention, the limited evidence available indicates that these additional features and design aspects of BFR apparatus also impact the acute psychophysiological responses to BFR exercise.[5] For example, Hughes *et al*[6] demonstrated that the use of BFR apparatus not designed to autoregulate pressure during dynamic exercise leads to spikes in interface pressure which may contribute to exacerbated acute psychophysiological responses to BFR exercise, for example, increases in perceived discomfort, exertion and haemodynamic responses. Furthermore, Jacobs *et al*[8] demonstrated that the presence of autoregulation capabilities significantly reduced delayed onset muscle soreness and perceptual responses during BFR resistance exercise compared with when the BFR exercise was performed using the same apparatus without autoregulation. However, this is not a universal finding as other studies using a different apparatus showed minimal differences in perceptual responses between autoregulated and non-autoregulated conditions,[9] highlighting that outcomes with autoregulation may be device-specific.

REPORTING RECOMMENDATIONS

It is clear that the BFR apparatus used and methodology employed has a significant impact on the BFR stimulus provided, and a growing body of evidence suggests that it also impacts the acute psychophysiological responses to BFR applications, which has potential implications for the safety, efficacy and feasibility of BFR interventions. However, a key barrier to advancement of this knowledge is that many important features of BFR apparatus and methodology employed are overlooked and under-reported in the scientific literature.[5] For instance, a recent systematic review and meta-analysis compared the magnitude of muscular adaptations between 'regulated' and 'unregulated' BFR apparatus.[10] Yet, the vast majority of the 81 included studies did not report key detail regarding the BFR apparatus features (eg, autoregulation capabilities) or provide supporting data and evidence of accuracy. This exemplifies a broader issue of missing

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Table 1 Reporting recommendations for research investigating BFR applications

Category		
BFR instrument apparatus capabilities	**Reporting element**	Recommendation
Manufacturer and model	Name and details	Essential
Method of pressure measurement	For example, automatic vs manual method for measurement of limb occlusion pressure for individualisation of pressure	Essential
Pressure regulation	For example, autoregulation vs non-autoregulation	Essential
Validity and reliability of limb occlusion pressure measurement	For example, test–retest reliability, inter-rater and intra-rater reliability for manual methods including handheld doppler ultrasound, plethysmography and pulse oximetry, if multiple assessors	Desirable
Tourniquet cuff properties		
Manufacturer and model	Name and details	Essential
Cuff selection	Criteria used to determine appropriate cuff size for each limb(s)	Essential
Cuff width	Measured width in mm, cm or inches	Essential
Material	All cuff materials	Essential
Type of internal bladder system	For example, single vs multichambered	Essential
Cuff Shape	For example, straight vs contoured	Desirable
Internal bladder length	Specify whether the bladder fully encompasses the limb or not	Desirable
BFR pressure prescription		
Limb occlusion pressure	Absolute pressure required to reach limb occlusion pressure, in mm Hg or equivalent	Essential
Posture used for measurement of limb occlusion pressure	Report posture (eg, supine vs sitting vs standing) and relevant instructions provided to participants	Essential
Timings of pressure application	Including: ► Point of inflation and deflation pre-exercise, during exercise and postexercise ► Duration of inflation ► Inflation/deflation cycle timing	Essential
Target vs actual pressure applied	Expected/known differences in interface/bladder pressure vs target pressure (where known) for a specific BFR apparatus	Desirable

BFR, blood flow restriction.

critical information regarding BFR apparatus design and methodology in the literature.[5] Failure to specifically identify key features of BFR apparatus and methodology hinders efforts for researchers to reproduce, comparatively evaluate and further investigate results and outcomes. Therefore, we have developed reporting recommendations and relevant rationales for research, based on current available evidence (table 1). Additionally, detailed rationales for each recommendation are provided to allow readers to explore the underlying reasoning behind these recommendations (online supplemental 1). As the BFR field strives to identify optimal protocols to ensure consistency, maximise safety and improve efficacy of BFR applications, thorough and transparent reporting of the BFR apparatus and methodology employed will be decisive in the success of these efforts to advance BFR research practice. We urge future researchers, practitioners and peer reviewers to consider these reporting recommendations in the design and publication of scientific data associated with BFR applications.

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